A Physics-Inspired Continuous-Relaxation Heuristic for the Euclidean Traveling Salesperson Problem

A Constrained Dynamical Formulation and Reference Prototype

Elsayed Mohamed

Abstract

This paper proposes an exploratory, deterministic heuristic for the Euclidean Traveling Salesperson Problem (TSP). The method represents each possible undirected edge by a continuous activation variable in [0, 1] and evolves these variables using four interacting mechanisms: (i) a gradual continuation of normalized edge costs, (ii) degree correction through dual "suction" potentials and an augmented penalty, (iii) a geometric penalty for pairs of candidate edges that cross, and (iv) a gradually strengthened double-well penalty that favors values near 0 or 1. A separate reconstruction stage converts the relaxed edge weights into a Hamiltonian cycle using multi-start greedy construction followed by 2-opt improvement.

The formulation is deliberately narrower than a claim of a new physical law or a general-purpose TSP solver. In particular, the cost continuation is not described as mass-preserving erosion; the crossing term acts on pairs of geometrically intersecting segments rather than endpoint distances; and the derivative of the binary penalty is used with the sign that pushes interior values away from 1/2. The method does not guarantee global optimality, convergence to a binary state, or elimination of disconnected subtours. A reproducible NumPy prototype is provided as a research starting point. No benchmark results are claimed in this paper. The next scientific step is controlled comparison against standard baselines on public TSP instances.

Keywords: Euclidean TSP; continuous relaxation; primal-dual dynamics; geometric crossing penalty; nonconvex regularization; 2-opt; heuristic optimization.

1. Introduction

The symmetric Euclidean TSP asks for a shortest closed tour that visits each of n cities exactly once. It is a classical combinatorial optimization problem with a substantial literature on exact methods and heuristics. High-quality practical methods, including Lin–Kernighan variants, use carefully engineered discrete neighborhood searches and are established comparison points for any proposed heuristic (Helsgaun, 2000; Applegate et al., 2006). TSPLIB provides a standard collection of instances for repeatable evaluation (Reinelt, 1991).

Biologically inspired network models provide useful motivation for adaptive edge weights. For example, Tero et al. (2010) model aspects of network formation by the slime mold *Physarum polycephalum*. Such work motivates the exploration of local, continuously updated edge variables, but it does not establish that a continuous network model solves the TSP or avoids its subtour constraints automatically.

This paper turns that broad inspiration into a testable heuristic with a deliberately explicit mathematical specification. Its main goal is methodological: define the state, penalties, dynamics, reconstruction procedure, and limitations clearly enough that independent researchers can reproduce or reject the approach.

The paper makes no claim to prove P = NP, to provide a polynomial-time algorithm with a guaranteed approximation ratio, or to preserve the global minimizer under smoothing. The continuous relaxation is an exploratory search mechanism; the discrete tour is produced by a separate reconstruction and local-search stage.

2. Problem Definition and State Variables

Let the cities be points p_i = (u_i, v_i) in the Euclidean plane, for i = 1, ..., n. Let L_ij = ||p_i - p_j||$_2$ be the Euclidean distance between cities i and j. We use the complete undirected graph with edge set

$$E = \{\{i, j\} : 1 \le i < j \le n\}.$$

Each edge e = {i, j} has a continuous activation x_e ∈ [0, 1]. The corresponding symmetric matrix X has X_ij = X_ji = x_{ij} for i ≠ j and X_ii = 0. This representation avoids duplicating the variables for the two orientations of an undirected edge.

The relaxed degree at city i is

$$d_i(x) = \Sigma_{j \neq i} x_{ij}, \quad r_i(x) = d_i(x) - 2.$$

A Hamiltonian cycle satisfies d_i = 2 for every city, but degree constraints alone are not sufficient: a 2-regular graph may be a disjoint union of multiple cycles. Consequently, the continuous state X is not itself treated as a valid tour, even when its row sums are close to 2.

For numerical stability, normalized edge costs are used:

$$\ell_{ij} = L_{ij} / \bar{L}, \quad \bar{L} = (1 / |E|) \Sigma_{e \in E} L_e.$$

Instances with $\bar{L}$ = 0 are degenerate and are rejected by the implementation.

3. The Four Mechanisms

3.1. Cost continuation (replacing the mass-preserving erosion claim)

The original “mass-preserving erosion” interpretation is not retained as a mathematical result. Gaussian convolution of a function and mass conservation of a measure do not, by themselves, imply preservation of the location or identity of its global minimizer. The earlier formulation did not define a probability measure or an optimal-transport map on the TSP objective, so a Wasserstein invariant cannot be inferred from it.

Instead, the prototype uses a transparent cost-continuation schedule. Let ρ(t) decrease from a chosen value ρ_0 in [0, 1) to zero. Define

$$\ell_e(t) = (1 - \rho(t))\, \ell_e + \rho(t),$$

where the normalized mean edge cost is one. Early in the run, this reduces differences between edge-cost coefficients in the relaxed dynamics; later, the original normalized distances are restored.

This schedule has a precise but limited property. Every Hamiltonian tour uses exactly n edges, so for any tour T,

Σ_{e ∈ T} ℓ_e(t) = (1 − Σ(t)) ρ_{e ∈ T} ℓ_e + ∈(t)n.

For ρ(t) < 1, the ranking of valid tours by length is therefore unchanged by this affine transformation. However, the trajectory of a continuous relaxation can change because intermediate states need not have exactly n total edge activation. This continuation is a heuristic numerical device, not a proof that local minima disappear or that a global optimizer remains in the same basin.

3.2. Degree correction and active suction

The degree target is two at every city. The method uses dual variables s_i as “suction potentials”: if city i is under-connected, s_i increases and makes incident edges more attractive in the primal update. The sign convention is fixed by the Lagrangian term −Σ_i s_i(d_i − 2).

A quadratic degree penalty is also included to resist large degree residuals during finite-step integration. This produces an augmented, nonconvex primal-dual heuristic. The combined use of a quadratic penalty and dual feedback is a stabilizing design choice; it does not guarantee that the degree equations are satisfied exactly at every iteration.

3.3. Geometric crossing penalty

Let P be the set of unordered pairs {e, f} of candidate edges that have four distinct endpoints and whose straight-line segments properly intersect. Define

$$C(x) = \Sigma_{\{e,f\} \in P} x_e x_f.$$

The gradient contribution for an edge e is

$$\partial C/\partial x_e = \Sigma_{f : \{e,f\} \in P} x_f.$$

This term penalizes simultaneous activation of crossing edges. It corrects a substantive mismatch in the earlier prototype: the Euclidean distance between two cities is not the distance between two candidate route segments, and a penalty based only on L_{ij} does not detect a crossing between segments (i, j) and (k, l).

The implementation uses a binary geometric crossing relation for fixed city coordinates. It does not use a differentiable near-crossing distance, and it does not guarantee that the final reconstructed tour is crossing-free. In a non-degenerate Euclidean TSP, a proper crossing can be removed by a 2-opt exchange that shortens the tour, so a crossing penalty is a plausible heuristic preference rather than a replacement for tour validation or local search.

3.4. Fractional crystallization via a double-well penalty

For an edge activation x_e, define

$$P_{freeze}(x_e) = \mu(t)\, x_e(1 - x_e),$$

where $\mu(t)$ increases from zero to μ_{max}. This penalty is zero at 0 and 1 and positive for values strictly between them. Its derivative is $\mu(t)(1 - 2x_e)$, so the negative-gradient contribution is

$$-\partial P_{freeze}/\partial x_e = \mu(t)(2x_e - 1).$$

The sign matters. For x_e below 1/2, this term pushes the variable downward; for x_e above 1/2, it pushes it upward. A term of the form $-\mu x(1-x)$ in the differential equation would instead drive all interior values downward and would not implement the stated double-well mechanism.

Crystallization makes the relaxation nonconvex. It does not guarantee that every variable reaches an exact endpoint, that the endpoints obey degree constraints, or that the selected edges form one Hamiltonian cycle. These properties must be checked or repaired by a separate discrete stage.

4. Coupled Primal-Dual Dynamics

Let $\gamma \geq 0$ weight the crossing penalty, $\beta \geq 0$ weight the quadratic degree penalty, and $\mu(t)$ be the time-varying crystallization strength. Define the time-dependent augmented Lagrangian

$$J_t(x, s) = \Sigma_e \ell_e(t) x_e + \gamma\, C(x) + \mu(t) \Sigma_e\, x_e(1-x_e) + (\beta/2)\Sigma_i\, r_i(x)^2 - \Sigma_i\, s_i\, r_i(x).$$

For $e = \{i, j\}$, differentiation gives

$$\partial J_t/\partial x_{ij} = \ell_{ij}(t) + \gamma \Sigma_{f : \{e,f\} \in P} x_f + \mu(t)(1-2x_{ij}) + \beta(r_i + r_j) - s_i - s_j.$$

The continuous model is the projected primal-descent / dual-ascent system

$$\dot{x}_{ij} = \Pi_{T([0,1](x_{ij}))}\, (-\partial J_t/\partial x_{ij}),$$

$$\dot{s}_i = \eta(2 - d_i(x)),$$

where $\eta > 0$ is a dual update rate and Π_T denotes projection onto the tangent cone of the interval [0, 1]. At a boundary, the projected flow cannot move x_{ij} outside the feasible interval. The reference code approximates this projected flow with explicit Euler steps followed by clipping to [0, 1].

Combining the terms yields the update direction used by the implementation:

$$\dot{x}_{ij} = -\ell_{ij} + s_i + s_j - \beta(r_i + r_j) - \gamma \Sigma_{f : \{e,f\} \in P} x_f + \mu(t)(2x_{ij} - 1).$$

The schedule decreases $\rho(t)$ toward zero and increases $\mu(t)$ toward μ_{max}. The ODE is nonconvex and coupled to dual variables. It is not established here that the complete primal-dual flow has a global Lyapunov function, converges for all initial conditions, or avoids limit cycles and step-size artifacts. We therefore make no general convergence theorem claim.

5. Discrete Tour Reconstruction

The output of the relaxation is a symmetric edge-weight matrix, not a Hamiltonian cycle. The reference implementation therefore has a distinct reconstruction stage:

1. Form adjusted edge scores $A_{ij} = L_{ij} - \kappa \bar{L} X_{ij}$, where $\kappa \geq 0$ controls the influence of relaxed affinity.
2. For each possible start city, construct a tour greedily by selecting the unvisited city with the smallest adjusted score.
3. Apply symmetric 2-opt local search using the original Euclidean distances, not the adjusted scores.
4. Return the shortest reconstructed tour among the multi-start candidates and verify that every city appears exactly once.

This reconstruction always returns a Hamiltonian cycle for $n \geq 3$ if it completes normally, because it constructs a permutation of all cities and closes the cycle implicitly. It does not guarantee an optimal cycle. The role of X is to bias candidate construction; the actual quality must be assessed by tour length against baselines.

6. Reference Algorithm and Reproducibility

The companion file `physics_tsp_reference.py` contains a self-contained NumPy implementation. It includes deterministic instance generation, Euclidean distance calculation, precomputation of proper crossing-edge pairs, the projected primal-dual updates, multi-start reconstruction, 2-opt, and basic validity checks. The script is intended for small and moderate instances, not for large-scale deployment.

A reproducible run should report at least: random seed or input instance identifier; n; all hyperparameters; number of integration steps; runtime; final tour length; relaxed degree residuals; number of crossing pairs precomputed; and whether the returned tour passed the permutation check. Hyperparameters should be selected on separate tuning instances, rather than tuned on the test set.

7. Analysis, Complexity, and Limits

7.1. What can be stated safely

The projection step guarantees $0 \leq x_e \leq 1$ at every stored iteration. Symmetric representation and a zero diagonal are guaranteed by construction. The reconstruction procedure generates a permutation of the cities and then treats it as a closed cycle. These are implementation invariants, not statements about optimality.

The degree feedback aims to drive d_i toward two. Even if the degree residuals become small, disconnected 2-regular components remain possible. A positive algebraic-connectivity value λ_2 of a nonnegative graph Laplacian is a connectivity indicator for the associated weighted graph, but connectivity alone does not establish Hamiltonian structure; this spectral idea is not implemented in the present prototype (Fiedler, 1973).

7.2. Complexity of the reference implementation

There are $m = n(n-1)/2$ candidate edges. Computing the full Euclidean distance matrix costs $O(n^2)$. The implementation checks pairs of candidate edges to identify geometric crossings, which costs $O(m^2) = O(n^4)$ in the worst case. Let $q = |P|$ be the number of crossing pairs. One relaxation step costs $O(m + q + n)$, and the crossing data require $O(q)$ storage. Since q can be $O(n^4)$ in the worst case, this reference implementation is not a scalable $O(n^2)$ or $O(n^3)$ solver as originally suggested.

For a fixed number K of relaxation steps, the worst-case operation count for this implementation is $O(K\, n^4)$, excluding implementation-dependent numeric costs. This observation is not a polynomial-time solution claim for the TSP in the complexity-theoretic sense: no approximation guarantee, exactness guarantee, or polynomial

bound on the steps needed to achieve a specified solution quality has been established. A complexity statement about a fixed number of iterations does not imply that the returned tour is good.

7.3. Empirical validation plan

No benchmark results are claimed here. An evaluation should compare the method with at least nearest neighbor, nearest neighbor plus 2-opt, and a strong implementation such as LKH; an exact solver can be added on small instances where optimal values are available. TSPLIB instances and controlled random Euclidean instances should be used, with multiple seeds where applicable.

Report tour length, relative gap to a known optimum or best-known value, runtime, run-to-run variability, and invalid-solution rate. Use ablations to remove each mechanism in turn: cost continuation, dual suction/degree penalty, crossing penalty, and crystallization. Parameters and stopping criteria must be fixed before the held-out test runs. All code, instance identifiers, and raw results should be published to make the comparison reproducible.

7.4. Explicit limitations

The method has no proof of global convergence or global optimality. Crystallization is a nonconvex heuristic penalty, not a proof that the integrality gap is closed. Crossing penalties and degree correction do not eliminate subtours. The dual variables can oscillate or drift under poor step sizes. The crossing-pair preprocessing has $O(n^4)$ worst-case cost. The included greedy-plus-2-opt reconstruction may dominate the final quality, so a proper ablation is necessary to measure whether the continuous relaxation adds value beyond the baseline.

8. Conclusion

This paper provides a more precise and falsifiable formulation of a physics-inspired continuous-relaxation heuristic for the Euclidean TSP. Its four components are normalized cost continuation, degree correction through dual suction and an augmented penalty, a geometric crossing penalty, and double-well crystallization. The formulation corrects the sign of the crystallization force, places repulsion on crossing route segments rather than endpoint distances, and explicitly separates a relaxed matrix from a valid tour.

The central open question is empirical: does the relaxed state contain useful information that leads to better tours than simple construction and local search alone? The current work does not answer that question because no benchmark study has been performed. The next contribution should be a controlled evaluation with baselines and ablations, not stronger convergence or complexity claims. The code is offered as a reproducible starting point for that evaluation.

References

Applegate, D. L., Bixby, R. E., Chvátal, V., and Cook, W. J. (2006). *The Traveling Salesman Problem: A Computational Study*. Princeton University Press.

Fiedler, M. (1973). Algebraic connectivity of graphs. *Czechoslovak Mathematical Journal*, 23(2), 298–305. https://doi.org/10.21136/CMJ.1973.101168

Helsgaun, K. (2000). An effective implementation of the Lin–Kernighan traveling salesman heuristic. *European Journal of Operational Research*, 126(1), 106–130. https://doi.org/10.1016/S0377-2217(99)00284-2

Reinelt, G. (1991). TSPLIB—A traveling salesman problem library. *ORSA Journal on Computing*, 3(4), 376–384. https://doi.org/10.1287/ijoc.3.4.376

Tero, A., Takagi, S., Saigusa, T., Ito, K., Bebber, D. P., Fricker, M. D., Yumiki, K., Kobayashi, R., and Nakagaki, T. (2010). Rules for biologically inspired adaptive network design. *Science*, 327(5964), 439–442. https://doi.org/10.1126/science.1177894

Peyré, G., and Cuturi, M. (2019). *Computational Optimal Transport*. Foundations and Trends in Machine Learning, 11(5–6), 355–602. https://doi.org/10.1561/2200000073

Appendix A. Reproducibility notes

Python 3.10+ and NumPy are required. Run `python physics_tsp_reference.py`. The script creates a deterministic random Euclidean instance with 15 cities, solves the relaxation, reconstructs a tour, prints the tour length and degree range, and marks the result as heuristic. This smoke test checks execution and tour validity; it is not a benchmark and does not demonstrate a solution-quality advantage.